CHAPTER QUESTIONS

SKILLS CRITICAL THINKING 6

1 a Alkanes are *saturated* hydrocarbons. Explain the term *saturated*.

b Undecane is an alkane with 11 carbon atoms.

7

i Write down the molecular formula for undecane.

SKILLS REASONING 5

ii What physical state (solid, liquid or gas) would you expect undecane to be in at room temperature (25 °C)?

SKILLS PROBLEM SOLVING 9

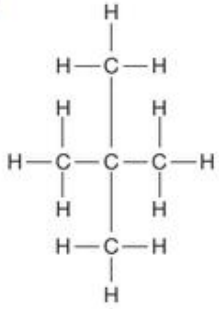
iii Write a balanced chemical equation for the complete combustion of undecane.

iv Write a balanced chemical equation for the incomplete combustion of undecane to produce carbon monoxide and explain why this reaction is harmful to humans.

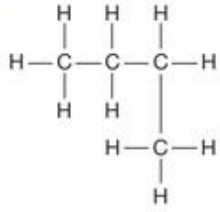
SKILLS ANALYSIS 7

2 Write the molecular formulae of the following compounds.

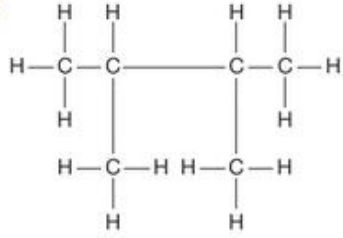
a



b



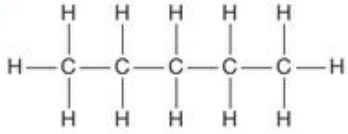
c



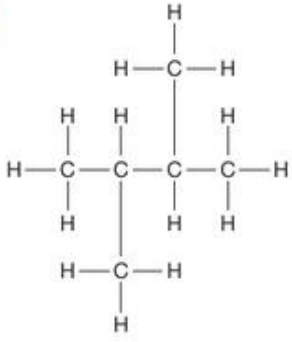
SKILLS INTERPRETATION

3 Draw two isomers of each of the following compounds.

a



b



SKILLS CRITICAL THINKING 5

4 Ethane reacts with bromine under certain conditions.

a State the conditions needed for this reaction to occur.

6

b What is the type of reaction that occurs?

SKILLS PROBLEM SOLVING 9

c Write a balanced chemical equation for the reaction using displayed formulae and name the organic product produced.

SKILLS INTERPRETATION

5 Butane reacts with chlorine to form two organic compounds with the formula C_4H_9Cl.

a Write a balanced chemical equation for the formation of one of these compounds.

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b Draw the structures and give the names of the two possible *organic* products of this reaction.